

Internship day-1

Report

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- **CIA Triad**

The **CIA** (Confidentiality, Integrity, Availability)

Triad is a foundational model in information security that defines the three core objectives an organization must protect to secure information assets. It is used to guide security policy design, risk assessment, and control selection.

1. Confidentiality

Objective: Ensure that information is accessible only to authorized individuals, systems, or processes.

Confidentiality prevents unauthorized disclosure of sensitive data such as personal information, intellectual property, financial records, and credentials.

Common threats

- Data breaches
- Insider threats
- Eavesdropping and packet sniffing
- Social engineering

2. Integrity

Objective: Ensure that data remains accurate, complete, and unaltered except by authorized means.

Integrity protects information from unauthorized modification, deletion, or corruption, whether accidental or malicious.

Common threats

- Malware altering data
- Unauthorized database changes

- Man-in-the-middle attacks
- Software bugs or human error

3. Availability

Objective: Ensure that information and systems are accessible to authorized users when needed.

Availability focuses on uptime, reliability, and timely access to systems and data.

Common threats

- Denial-of-Service (DoS/DDoS) attacks
- Hardware failures
- Power outages
- Natural disasters

• Types of Cyber Attacks

Cyber attacks are deliberate attempts to compromise **Confidentiality, Integrity, or Availability** of information systems. Below are the most common types, explained concisely.

1. Malware Attacks

Malicious software designed to damage, disrupt, or gain unauthorized access to systems.

Examples: Virus, Worm, Trojan, Ransomware, Spyware

Impact: Data theft, system damage, unauthorized control

2. Phishing Attacks

Fraudulent messages (email, SMS, calls) that trick users into revealing sensitive information.

Types: Email phishing, Spear phishing, Whaling

Impact: Credential theft, financial fraud

3. Denial of Service (DoS) / Distributed DoS (DDoS)

Overwhelming a system or network with traffic to make it unavailable to users.

Impact: Service downtime, business disruption

4. Man-in-the-Middle (MITM) Attacks

An attacker secretly intercepts and possibly alters communication between two parties.

Common scenarios: Public Wi-Fi attacks, session hijacking

Impact: Data theft, unauthorized transactions

5. SQL Injection

Malicious SQL queries are inserted into input fields to manipulate databases.

Impact: Data leakage, data deletion, unauthorized admin access

6. Cross-Site Scripting (XSS)

Injection of malicious scripts into trusted websites, executed in users' browsers.

Impact: Session hijacking, credential theft

7. Password Attacks

Attempts to crack or steal passwords.

Methods: Brute force, Dictionary attack, Credential stuffing

Impact: Account takeover

8. Zero-Day Attacks

Exploitation of unknown software vulnerabilities before patches are available.

Impact: Severe system compromise

9. Social Engineering Attacks

Psychological manipulation to trick people into breaking security policies.

Examples: Pretexting, Baiting, Tailgating

Impact: Unauthorized access, data theft

10. Advanced Persistent Threats (APT)

Long-term, targeted attacks by skilled attackers, often nation-state or organized groups.

Impact: Espionage, large-scale data exfiltration

• Attack Surface

Attack Surface is the collection of all entry points, vulnerabilities, interfaces, and exposures that an attacker could use to compromise a system

Main Types of Attack Surfaces

1. Digital Attack Surface

All software- and network-based exposure points.

Examples:

- Web applications and APIs
- Open ports and services

- Operating systems
- Databases
- Cloud services
- IoT devices

Risk: Exploitation through malware, injection attacks, misconfigurations.

2. Physical Attack Surface

Physical access points that attackers can exploit.

Examples:

- Unlocked servers
- USB ports
- Laptops and mobile devices
- Data centers

Risk: Hardware tampering, data theft, device compromise.

3. Human Attack Surface

People who can be manipulated or make mistakes.

Examples:

- Employees
- Contractors
- Third-party vendors

Risk: Phishing, social engineering, insider threats.

Common Components of an Attack Surface

- User input fields
- Network interfaces
- Authentication mechanisms
- Software vulnerabilities
- Third-party integrations
- Misconfigured permissions